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306 N Wright St.
Urbana, IL, 61801
Dept. of Electrical & Computer Engineering
University of Illinois Urbana-Champaign

EDUCATION

University of Illinois Urbana-Champaign

Aug. 2023 – Present

B.S. in Computer Engineering, GPA: 3.97/4.00

Zhejiang University (Dual Degree Program)

Aug. 2023 – Present

B.S. in Computer Engineering, GPA: 4.20/4.30

Selected Courseworks (all A/A+):

ECE 408 Applied Parallel Programming, ECE 391 Computer Systems Engineering

SKILLS

Programming: Python, C/C++, OpenAI Triton, CUDA C/C++, cuTile, Java, RISC-V Assembly

Frameworks: PyTorch, vLLM, SGLang, SkyRL, DeepSpeed, Flash-Linear-Attention, OpenHands

Tools: Git, Linux/UNIX, slurm, Docker, GDB, PDB, Azure, AWS, Google Cloud (GCP)

EXPERIENCE

TileGym (NVIDIA’s cuTile Kernel Library) [✉](#)

Dec. 2025 - Present

Open-Source Code Contributor

Remote

- Implemented backward pass optimization for FlashAttention and SwiGLU in cuTile, achieving more than 10% higher throughput than OpenAI’s kernel team’s implementation in Triton official tutorial.

University of Illinois Urbana-Champaign

Aug. 2025 – Present

Machine Learning Systems Researcher

Urbana, IL

ThunderAgent: A Fast, Simple, and Robust Program-Aware Agentic Inference System [✉](#)

- Co-first author paper in submission to ICML 2026, advised by Simran Arora at Together AI
- Proposed a program abstraction that wraps stateless LLM requests into stateful, schedulable programs, enabling lifecycle-aware scheduling of GPU KV caches, tool environments (Docker containers, sandboxes), and cross-node routing for distributed agentic inference and RL rollout across vLLM and SGLang backends.
- Designed a Periodic Thrashing Monitor with shortest-first eviction and exponential time-decay scoring to preemptively detect KV-cache thrashing, and a Global Waiting Queue with load-balanced cross-node restore that eliminates memory imbalance across multi-GPU nodes without sacrificing cache locality.
- Achieved 1.5–3.6× serving throughput and 1.8–3.9× RL rollout throughput on agentic workloads (OpenHands, mini-SWEAgent, ToolOrchestra) on 8×H100 nodes. Lifecycle-aware garbage collection yielded 4.2× disk memory savings by automatically reclaiming zombie tool resources upon program termination.

Zhejiang University

Aug. 2023 - Jul. 2025

Machine Learning Systems Researcher

Hangzhou, China

AuroraLong: Bringing RNNs Back to Efficient Open-Ended Video Understanding [✉](#)

- First author paper that pioneered the use of scalable Linear Attention to unlock hour-long video understanding on edge GPUs with limited VRAM, accelerated inference times by **8.13x** on RTX 3090 GPU.

Video-MMLU: A Massive Multi-Discipline Lecture Understanding Benchmark [✉](#)

- Developed a benchmark to evaluate reasoning capabilities of multimodal LLMs, earning Oral Presentation and the Outstanding Paper Award at Knowledge-Intensive Multimodal Reasoning Workshop, ICCV 2025.

Zhejiang University - University of Illinois Urbana-Champaign Institute

Aug. 2024 – May 2025

Undergraduate Teaching Assistant

Hangzhou, China

- **ECE 220 (Computer Systems & Programming)**

Spring 2025

- **ECE 120 (Introduction to Computing)**

Fall 2024